

RideWise: Customer Analytics & Churn Prediction

This presentation outlines our 3-week sprint project to develop a robust customer analytics platform for RideWise, focusing on customer segmentation and churn prediction.

Our goal is to leverage data-driven insights to significantly reduce customer churn and enhance retention strategies across key European markets.



RideWise

Customer Segmentation & Churn Prediction Engine

3-Week Sprint Project



Company Overview & Business Context

- 1** **RideWise Company Profile**
RideWise is a leading European mobility technology company serving over 200,000 customers and powering more than 800,000 monthly trips across major cities.

- 2** **Industry Context**
RideWise operates in the fast-paced European mobility and transportation technology sector, providing innovative solutions to address the evolving urban transit needs.

- 3** **Current Scale & Market Position**
With a strong operational footprint and rapidly growing customer base, RideWise is well-positioned as a market leader in the European mobility technology landscape.

Business Challenge & Problem Statement

CURRENT STATE



CHURN FUNNEL

DESIRED STATE



TARGET

Current State: Blind Spots & Lost Revenue

- High Churn: 25% quarterly churn, eroding user base and revenue.
- Limited Insight: Lack of real-time understanding of customer behavior.
- Ineffective Marketing: Poorly targeted campaigns, wasted spend.
- Reactive Approach: No proactive identification of at-risk users.
- Slow Analysis: Manual processing delays insights, hindering decisions.

Desired State: Proactive Retention

- Enhanced Retention: Proactive identification and action on at-risk customers.
- Optimized Spend: Targeted campaigns via deep customer segmentation.
- Competitive Edge: Proactive churn mitigation for market leadership.
- Real-time Insights: Automated, instant, actionable insights.

Project Rationale & Objectives

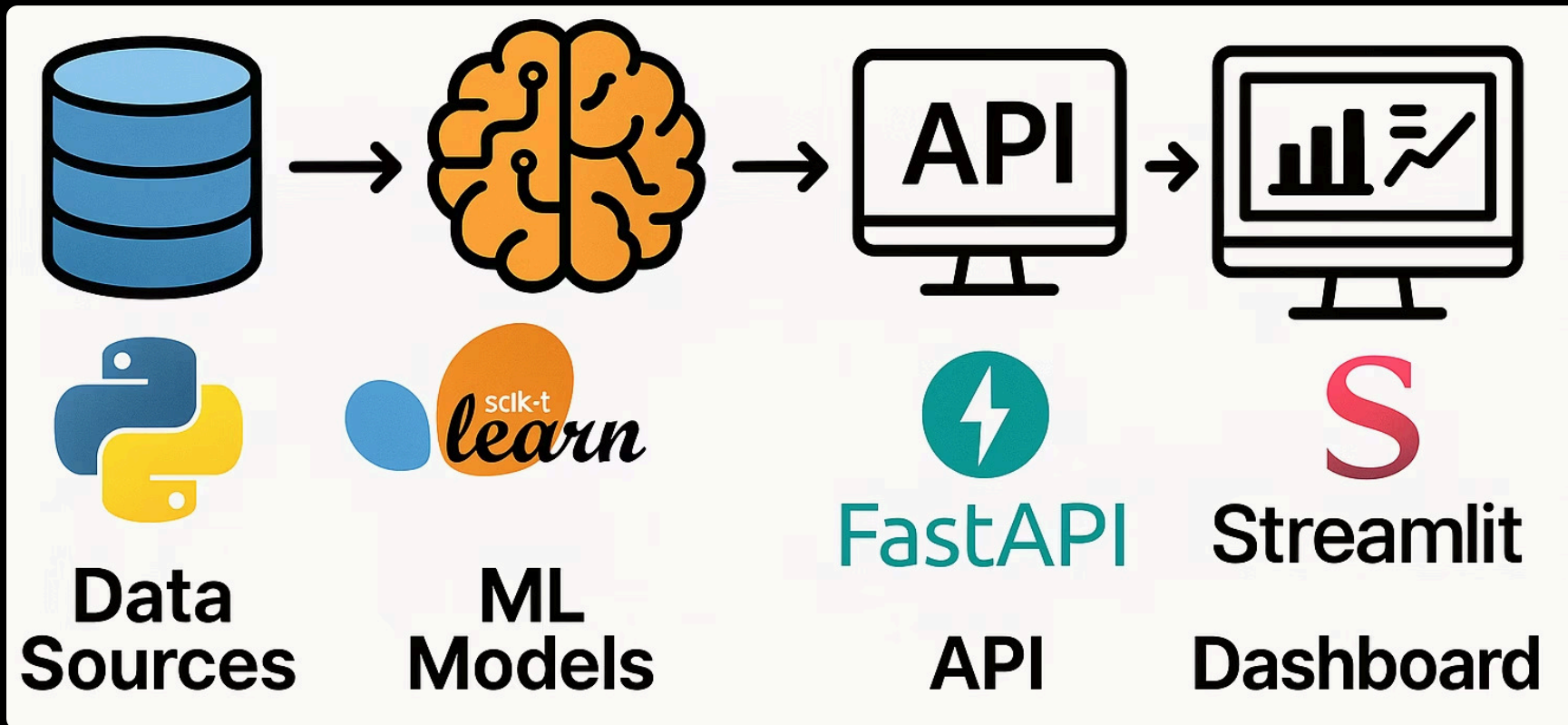
Clear Project Objectives

- **Churn Prediction:** Develop a robust model to identify at-risk users before they leave.
- **Customer Segmentation:** Create meaningful segments to understand diverse user behaviors.
- **Real-time Analytics:** Provide instant, data-driven insights for rapid decision-making.

Expected Business Outcomes

- **Reduced Churn:** Significantly lower customer attrition rates, safeguarding revenue.
- **Increased LTV:** Boost Customer Lifetime Value through targeted retention strategies.
- **Optimized Marketing:** More effective and personalized campaigns, maximizing ROI.
- **Enhanced Competitiveness:** Strengthen RideWise's market position through proactive data-driven actions.

Solution Architecture: From Data to Decision



Core Components

- **Data Pipeline:** Ingests customer, trip, and behavioral data.
- **ML Models:** Powers churn prediction and customer segmentation.
- **Real-time API:** Provides instant customer risk scoring.
- **Analytics Dashboard:** Visualizes business insights and KPIs.

Technology Stack

- **Python:** Core language for data processing and modeling.
- **scikit-learn, pandas:** Libraries for ML and data manipulation.
- **FastAPI:** For high-performance API development.
- **SQLite, Streamlit:** Database and interactive dashboard.

Dataset & Data Model: The Foundation of Insight

1

Customers

200,000 unique profiles with demographics and signup details.

2

Trips

Over 800,000 monthly ride records including journey paths and payment information.

3

Behavior

Monthly aggregated metrics derived from RFM analysis.

4

Promotions

Campaign data with detailed usage tracking for effectiveness.

5

External Data

Integrated weather, holiday, and competitor activity data.

Methodology - Customer Segmentation

We identified three key customer segments to tailor retention strategies effectively.



Regular Commuters (30%)

Consistent weekday usage, predictable patterns. High retention potential, ideal candidates for premium services and loyalty programs.



Occasional Users (40%)

Infrequent usage, representing the highest churn risk. Highly price-sensitive and responsive to targeted promotions.



Weekend Users (30%)

Primarily use RideWise for entertainment and leisure. Exhibit seasonal patterns and respond well to event-driven promotions.

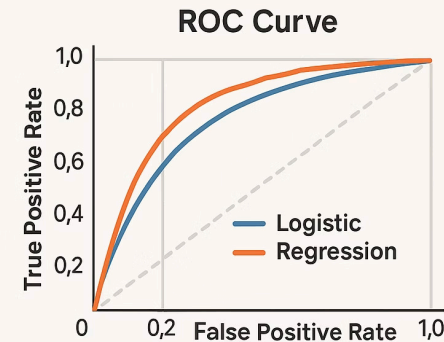
Methodology - Churn Prediction

Model Approach

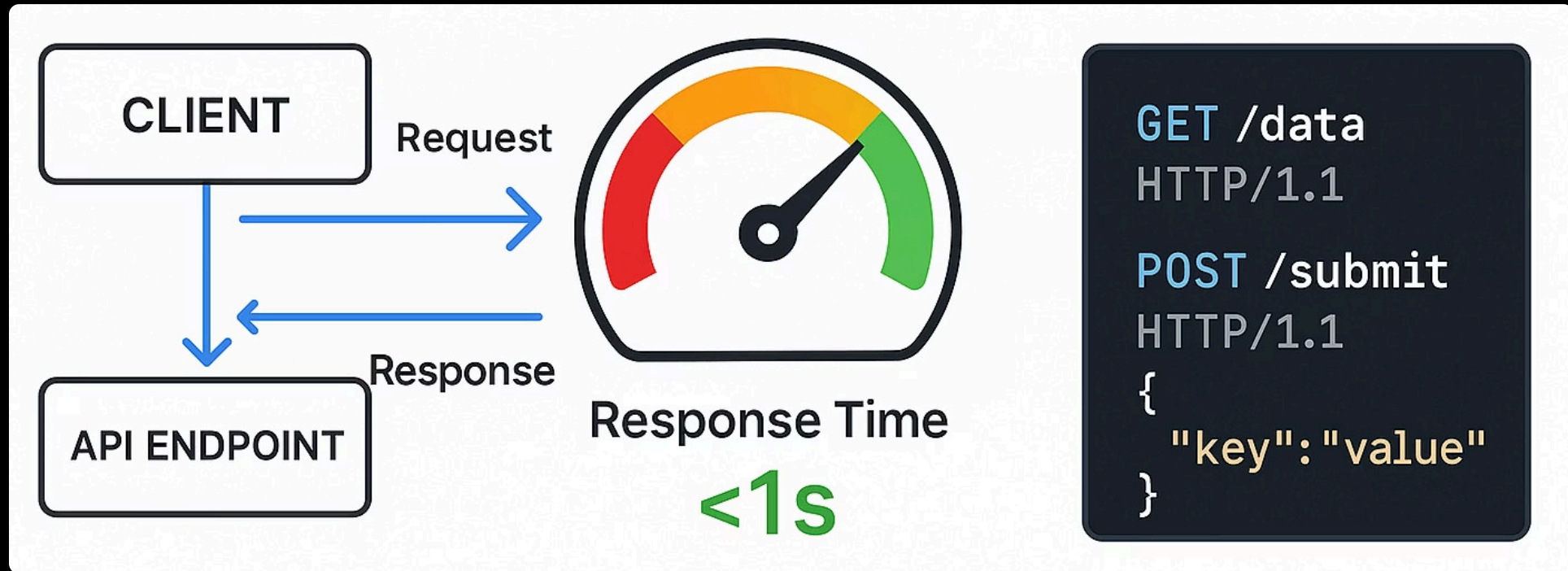
- **Primary Model:** Random Forest for its superior accuracy in complex datasets.
- **Baseline Model:** Logistic Regression for interpretability and benchmarking.
- **Target:** Predict 30-day inactivity risk to enable proactive engagement.

Model Comparison

	Logistic Regression	Random Forest
Accuracy	0,82	0,87
ROC AUC	0,86	0,92
F1 Score	0,74	0,79



Technical Implementation: API & Real-time Scoring



/predict/churn

Returns the probability of churn for a given customer.

/customer/segment

Assigns a customer to their predicted segment.

/dashboard/metrics

Aggregates and delivers key business performance indicators.

/health

Provides system status and uptime information.

Success Metrics & Expected Outcomes

Technical Success Metrics



Model Accuracy

AUC > **0.80** for churn prediction, ensuring high reliability.



Real-time Response

< **1 second** API response time for instant insights.



System Availability

99% uptime of the prediction API and dashboard.

Business Impact



Churn Reduction

Reduce **25%** quarterly customer churn rate.



Promotion Effectiveness

Improve the ROI and conversion rate of marketing campaigns.



Increased LTV

Boost Customer Lifetime Value through targeted engagement.